1) Imagine two polynomial regression models

A) $y = 2 + 0.8x + 0.05x^2$

B) $y = 2 + 15x - 94x^2 + 370x^3 - 650x^4 + 410x^5$

Suppose fit the training data equally well.

Ignoring the training data completely, which model would you trust more on new data, and why?

A, for several reasons. First is Occrum's Razor, saying all else being equal we should accept the simpler model. Also B has coefficients that are much larger, so it's going to be more exagerated, and likely over-fit. Model A's coefficients are small, so it's essentially doing less work to explain the data. It's less likely to magnify noise to the level of the data.

2) Why might a polynomial with coefficients like

300, -800, 1200, -700

be especially sensitive to tiny changes in the input?

Our model is stretching out the space to such a degree that the noise is going to be amplified too.

Even if we just divide all coefficients by 100, we still have a range of -8—12, we still have a range over an order of magnitude

3) What do you think happens when we gradually increase λ from 0 to 1000

What happens to Loss = prediction error + λ × penalty

- training error
- test error
- coefficient sizes?

At zero, our model is identical to least squares,

Our training error may increase, as we are forcing the model to use smaller and smaller parameters. Our test error may increase up to a point. Higher weight to small coefficients may hit a point where the coefficient size penalty is so large that it swamps the prediction error, and all we are really optimizing for is small coefficients.

4) Suppose one model has weights

(2,2,2,2) A

and another has weights

(8,0,0,0) B

Without calculating, which model do you think

- L2 dislikes more
- L1 dislikes more

L2 will dislike B more, since squaring larger numbers gets magnified more

L1 will dislike A more, since the sum of the errors will be higher

6.1) Which polynomial from Part 1 would you trust more?

A. My answers above still hold.

6.2) Why are very large coefficients geometrically unstable?

Noise is stretched as much as the data, and the path describing the data will oscilate a lot more,

6.3) As λ is increased, what happens to

- training error?
- test error?
- coefficients?

See ③ for more detail, but

- training error increases
- test error decreases
- coefficient sizes decrease

Up to a point where λ is so large that the data becomes irrelevant.

4) Where... oh... these are just reframing of the questions in the middle of the lesson. See ① above.

L1 likes values equal to zero
L2 likes smaller values